Smart Farming (ThingSpeak Channel 80502)

Models / AIntl Pipeline Evaluation — Task 2 Report

Branch: **kim/report/farming** (experiment only — not merged into **main**) | Location: **data_science/datasets/farming/**

Executive Summary

This report evaluates how the current Models/AIntl pipeline (*Dataset -> Models input validation -> Isolation Forest runtime -> Models adapter -> Correlation path -> Correlation adapter -> Analytics response -> Draft V0.1 validation*) behaves against a real, independently-sourced IoT dataset: a public ThingSpeak greenhouse climate-monitoring channel. No changes were made to the Models, Correlation, or AIntl implementation code — this is a behavioural evaluation of the pipeline as it stands.

- **7,261 readings** processed end-to-end through the full AIntl pipeline in **1.31s**, producing a Draft-V0.1-validated response with **535 alerts** (357 point anomalies, 178 correlation-change alerts).
- Isolation Forest flagged **357 of 7,261 readings (4.92%)** as anomalous on the air-temperature channel. Flags were concentrated on real, one-off excursions (a multi-hour heat spike and a separate cold dip), not the recurring day/night cycle, and were **100% deterministic** across repeated runs.
- The Correlation module produced a real but **unexpected** contrast: the deliberately weaker pairing (temp_c vs co2_ppm, 258 alerts) triggered **more** alerts than the genuinely physically-linked pairing (temp_c vs humidity_pct, 178 alerts) — because a near-constant, floored channel makes rolling Pearson correlation numerically unstable, a failure mode the module does not currently distinguish from a genuine relationship change.
- This dataset has **no ground-truth anomaly labels**; per the evaluation brief, no precision/recall/F1/AUC is reported. Evaluation instead relies on independent statistical cross-checks, determinism, and domain plausibility, detailed below.

Dataset Source and Provenance

- **Source:** ThingSpeak public channel 80502 (MathWorks ThingSpeak IoT platform), https://thingspeak.com/channels/80502 — retrieved via the public feeds.csv endpoint, no API key required.
- **What it is:** greenhouse climate-computer readings logged roughly once a minute: air temperature, a second temperature probe, relative humidity, and CO2, plus four derived psychrometric quantities (wet-bulb temperature, absolute humidity, dew point, humidity deficit).
- **Coverage:** 27 June – 12 October 2019, arriving as three disjoint export blocks separated by two multi-week outages (47 and 55 days).
- **Field identification (Task 1):** ThingSpeak exports only field1..field8 with no labels. The mapping used here was derived from the data itself and verified numerically by recomputing published psychrometric formulas from temp_c/humidity_pct and matching them to the file (errors at the 0.1 rounding resolution of the source data). This confirmed 4 of the 8 fields (wet_bulb_c, abs_humidity_gm3, dew_point_c, humidity_deficit_gm3) are deterministic functions of temperature and humidity, not independent sensor readings, and they were excluded from correlation analysis on that basis.

Transformations applied before the pipeline run

- Renamed field1..field8 to their identified names.
- Removed 22 rows (0.29% of the file) where every field reports a fixed error-code sentinel (99.9 / 455 / 9999 / 0.0) instead of a real reading.
- Selected the single longest continuous export block (7-12 Oct 2019, 7,261 rows at ~60s cadence) as the working series, to avoid two multi-week outages sitting inside rolling correlation windows.

- Formatted created_at as an ISO-8601 string, the format analytics_integration.pipeline expects.
- No imputation, resampling, or scaling was required beyond this.

Variable choice

- **Models anomaly-detection metric:** temp_c (air temperature).
- **Correlation pair A (primary):** temp_c vs humidity_pct — physically forced to move in opposite directions (warmer air holds more moisture), giving a known-sign relationship to test.
- **Correlation pair B (contrast):** temp_c vs co2_ppm — deliberately weaker, since the CO2 channel sits at its ~400ppm outdoor-baseline floor for most of the record.

Methodology

The pipeline was run exactly as implemented in analytics_integration.pipeline.run_analytics_pipeline, calling in turn the Models path (input validator -> IsolationForestDetector -> Models adapter) and the Correlation path (Correlation Flask service -> Correlation adapter), before both are combined into a single envelope and checked against the Draft V0.1 response contract. Configuration used throughout:

Parameter	Value
entity_id	greenhouse_ch80502
model_metric	temp_c
correlation_streams	[temp_c, humidity_pct] and [temp_c, co2_ppm]
detector_name	isolationforest
detector_parameters	{contamination: 0.05}
correlation_window_size / step_size	20 / 10
correlation_method	pearson

Data Integrity / EDA

- 7,488 raw readings x 10 columns (created_at, entry_id, field1..field8); all sensor fields numeric (float64).
- Zero duplicate rows, zero duplicate timestamps, zero NaN values anywhere in the raw file.
- That apparent cleanliness is misleading: 22 rows (0.29%) encode sensor failure as fixed error-code *values*, not NaN — invisible to any missing-value check.
- co2_ppm is heavily right-skewed with a hard floor at ~400ppm; temp_c/temp2_c are roughly bell-shaped; humidity_pct is wider and slightly left-skewed.

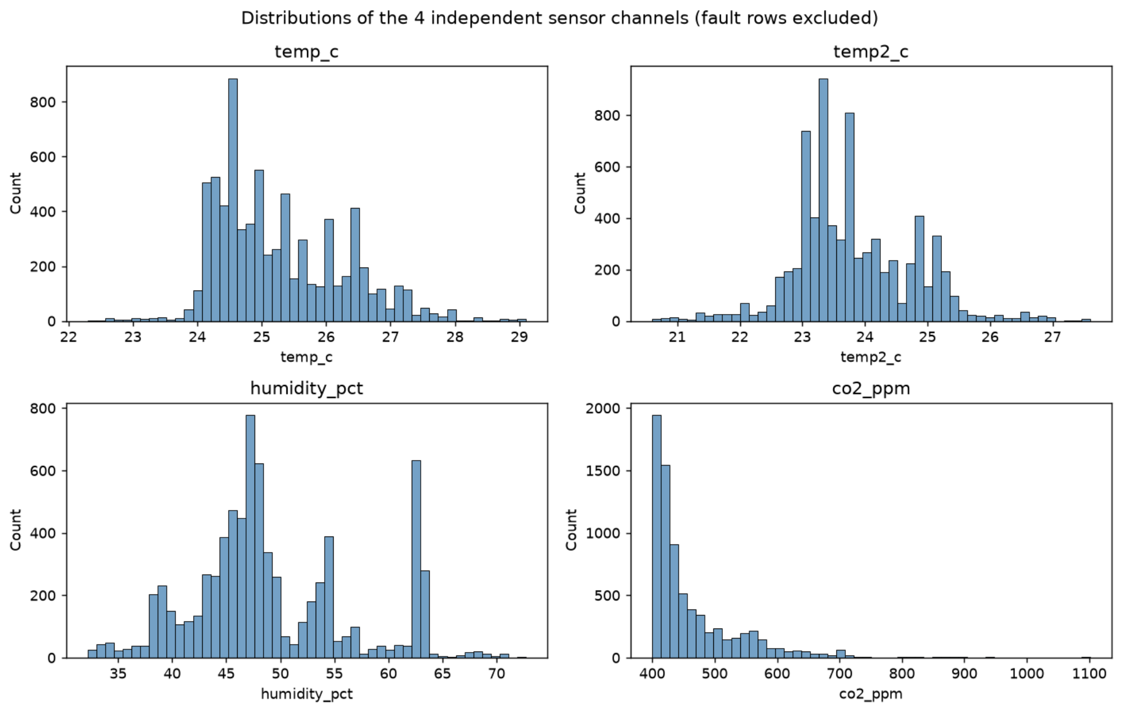

Figure 1 — Distributions of the 4 independent sensor channels (fault rows excluded).

Evidence: What the Input Validator Actually Catches

Three checks were run against the unmodified pipeline:

- **Missing field mapping** -> correctly rejected. Calling the pipeline with model_metric="temp_c" against the raw, unmapped file (which only has field1..field8) raises ValueError: Input data is missing required columns.
- **Duplicate timestamps / NaN sensor values** -> correctly rejected, exactly as documented, with clear InputValidationError messages.
- **Fault-sentinel values (99.9 / 9999 / 455 / 0.0)** -> not caught. These are valid, non-null floats, so the validator accepts them without complaint. The practical risk was limited on this run (the real fault block fell entirely in a block already excluded for other reasons), but a sentinel value chosen inside a sensor's normal operating range would pass through completely undetected. This is a gap in the validator's contract worth raising, not a defect introduced by this run.

Models Path — Isolation Forest on temp_c

- **357 / 7,261 readings (4.92%)** flagged, in line with the configured 5% contamination.
- Models runtime: **134.4 ms** for the full 7,261-row series.
- Flags were concentrated on 4 of 5 calendar days, with 2 of 5 days (7 and 11 Oct) receiving **zero** flags and 241 of 357 flags falling on a single day (9 Oct). That day's cluster lines up with a real multi-hour temperature spike to 29.1°C; a second, smaller cluster on 10 Oct lines up with a separate cold dip to ~22.3-22.6°C (see Figure 2). A detector mistaking the recurring day/night cycle for anomalous would instead flag similar counts at the same hour on every day — which is not what happened.
- **100%** of independently-identified top/bottom 0.5th-percentile temp_c readings were also flagged by the detector, and repeated runs were **bit-for-bit deterministic**.

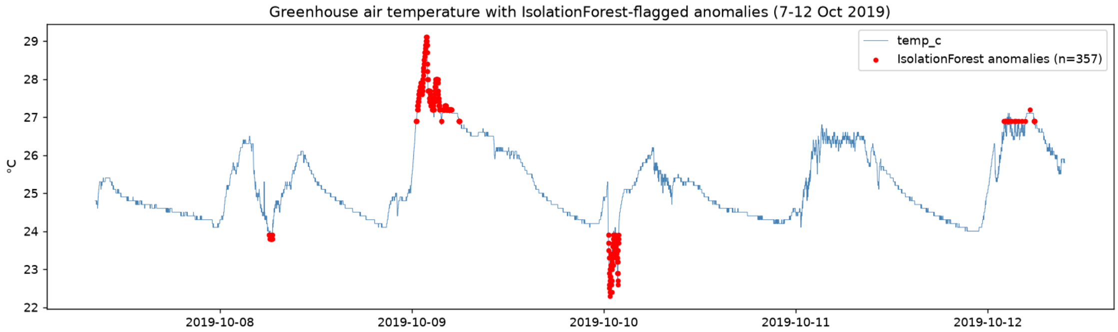

Figure 2 — Air temperature with IsolationForest-flagged anomalies (7-12 Oct 2019). Flags cluster on the real heat spike (9 Oct) and cold dip (10 Oct), not the recurring daily cycle.

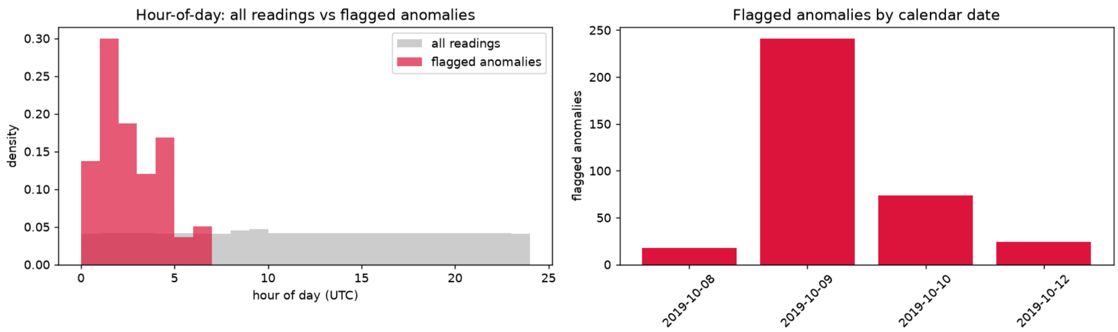

Figure 3 — Left: hour-of-day distribution of flagged anomalies vs all readings. Right: flagged anomalies by calendar date — 2 of 5 days received zero flags.

Correlation Path — Two Experiments

- **temp_c vs humidity_pct** (primary, physically-linked pair): **178 alerts** (46 HIGH / 65 MEDIUM / 67 LOW). Overall Pearson r = -0.569, matching the expected negative sign.
- **temp_c vs co2_ppm** (deliberately weaker pair): **258 alerts** (108 HIGH / 74 MEDIUM / 76 LOW).
- **This is the reverse of what was expected** — the weaker pair produced more alerts, and more severe ones. The mechanism: co2_ppm sits flat at its sensor floor for long stretches, so a 20-reading window covering one of those stretches has almost no variance in that channel — a few ppm of sensor jitter then dominates the Pearson coefficient, swinging it erratically between roughly +1 and -1. That is numerical instability from a near-constant channel, not a genuine drifting relationship, but the Correlation module currently reports both identically as CORRELATION_CHANGE alerts.

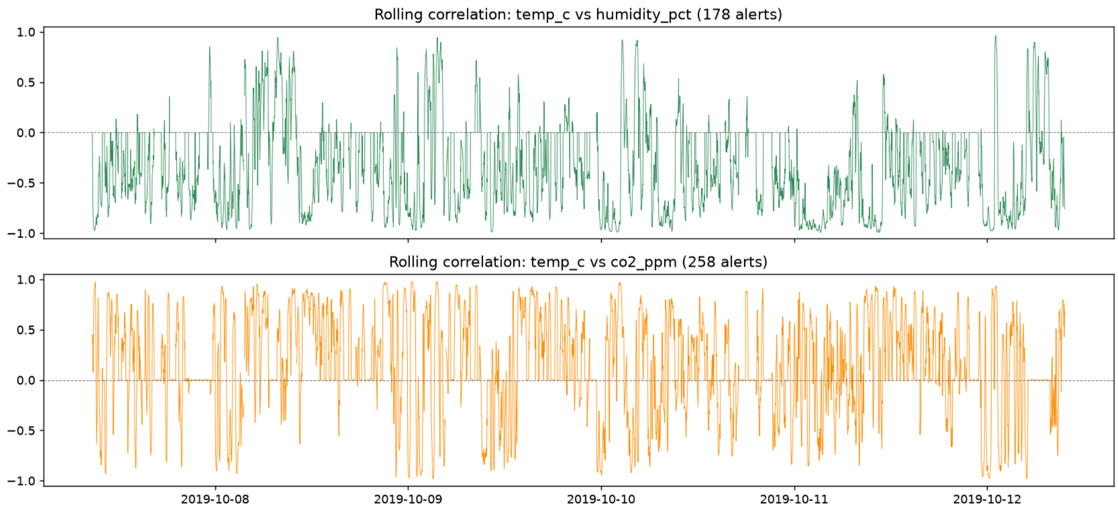

Figure 4 — Rolling correlation for both pairs. The co2 pairing (bottom) oscillates far more erratically than the humidity pairing (top), consistent with variance-floor instability rather than a real relationship.

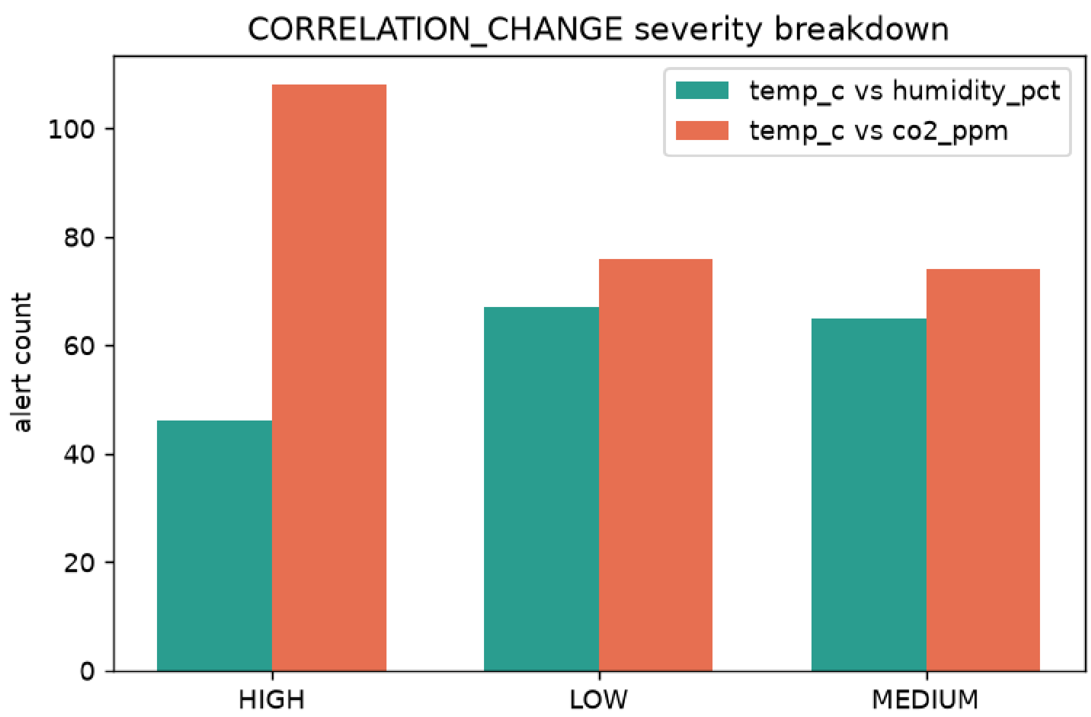

Figure 5 — CORRELATION_CHANGE severity breakdown: the weaker, floored pairing produced more HIGH-severity alerts than the genuinely physically-linked pairing.

Full AIntl Pipeline and Runtime

- The complete path — Models, Correlation, envelope building, and Draft V0.1 response validation — ran end-to-end with no code changes and completed in **1.31s** for the full 7,261-row block, producing **535 total alerts** (357 POINTWISE_ANOMALY, 178 CORRELATION_CHANGE).
- Runtime scaled close to linearly with row count from 300 to 7,261 rows, with no errors or warnings at any scale tested.

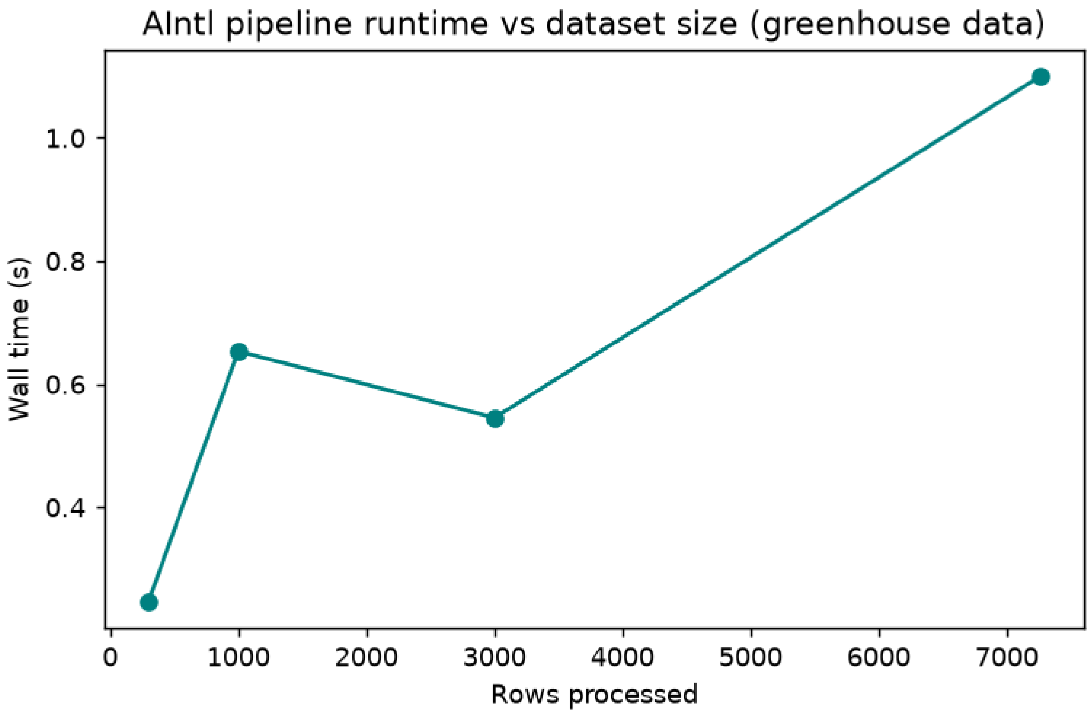

Figure 6 — AIntl pipeline wall-clock time vs. rows processed.

Limitations

- **No ground-truth labels** — no precision/recall/F1/AUC is reported; evaluation is qualitative (anomaly rate, timing plausibility, an independent statistical cross-check, determinism, and alert behaviour under two contrasting correlation pairs).
- **The input validator does not recognise domain-specific fault sentinels** — only literal NaN. A sentinel value inside a sensor's normal operating range would go undetected.
- **Neither the validator nor the correlation window is gap-aware** — a rolling window is defined by row count, not elapsed time, so it can silently straddle a real outage if that outage is not removed upstream by hand, as it was here.
- **The Correlation module cannot distinguish real drift from noise-driven instability** in a near-constant channel, as demonstrated directly by the co2 experiment above — a variance-floor guard would help, since floored/near-constant channels (a sensor at its detection limit, a valve permanently closed) are common in real IoT deployments.
- **temp2_c's physical location is undocumented** (soil, root zone, or elsewhere in the air).
- Single device, single 5-day working block — no cross-device or cross-season comparison possible.
- The Models path was exercised univariately (one channel at a time), not across multiple channels jointly.

Conclusions and Recommendations

This experiment provides further evidence that the Models/AIntl MVP generalises beyond its original NAB-derived development data to a second, independently-sourced, real IoT deployment with a materially different character: a strong daily periodicity, a live (not synthetic) sensor fault, and a genuinely physically-linked pair of channels. Preprocessing required was light — fault-sentinel removal and block selection, no imputation — and the full pipeline completed in well under two seconds for the whole usable block.

Recommended follow-ups, in priority order:

- Add a variance-floor guard to the Correlation module so a near-constant channel does not generate high-severity alerts purely from numerical instability.
- Document the input validator's fault-sentinel blind spot as a known boundary (NaN-only) rather than an assumed one, for teams integrating ThingSpeak-style devices that encode faults as sentinel values.
- Consider gap-awareness (elapsed time, not row count) for the rolling-correlation window.
- Extend the Models path to multivariate anomaly detection across the independent channels together.

Full reproducible analysis, code, and figures: data_science/datasets/farming/notebook.ipynb on branch kim/report/farming.